Federal Survey Question Consolidation

DRAFT — not validated — see disclaimer slide

Brock Webb

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Disclaimer

Warning

DRAFT — These materials are in development and have not been independently validated. The author makes no claims regarding validity until formal review is complete.

The views expressed are the author's own and do not necessarily represent the views of the U.S. Census Bureau or the U.S. Department of Commerce.

This is exploratory research assessing feasibility of AI-assisted survey analysis methods.

The Problem

- Federal surveys ask overlapping questions
- Respondent burden + data silos
- **This report:** Capstone analysis — which questions can actually consolidate?

Why It Matters

For Respondents: - Declining survey response rates + increasing costs - Overlapping questions create unnecessary burden - Consolidation reduces redundant questions

For Federal Agencies: - **Expert time is the scarce resource**, not expert capability - Traditional approach: weeks/months of manual comparison - **Our goal:** Multiply expert effectiveness → faster burden reduction

The Challenge

- Determining consolidation **requires expert judgment** (this doesn't change)
- Traditional timeline: **weeks to months** for manual review of 1,598 pairs
- Our approach: **AI handles pre-work, experts focus on highest-value decisions**
- Result: Structured expert review table ready for validation

What Experts Get

What AI Provides	What Experts Do
Pre-screened pairs with best matches	Validate match quality
Initial F1/F2/F3 classification	Confirm or override
Documented reasoning	Understand and critique
Confidence scores	Prioritize review effort
Barrier codes	Verify diagnosis

76% classified with high confidence $\rightarrow$ spot-check **24%** flagged for expert review $\rightarrow$ where judgment adds most value

Starting Point

- **6,987** survey questions across **47** federal survey instruments
- Mapped to Census Bureau standard taxonomy
- [Report 1 established concept classification]

Concept Classification (Report 1)

- LLM ensemble with arbitrator pattern
- Each question $\rightarrow$ best concept match
- Taxonomy: Demographics, Economics, Education, Health, etc.

Key Insight

! Important

Same concept   swappable data

- "Income" questions can measure completely different constructs
- Demographics (age, sex, race) = expected overlap (minority of questions)
- **Specialized content** = where real analysis matters

Harmonization Framework

Adopted standard survey harmonization classification:

Level	Definition
F1	Directly consolidable — no transformation needed
F2	Consolidable with transformation — needs recoding
F3	Not consolidable — fundamental barriers

Barrier codes: CC (construct/concept), TC (temporal), RS (response scale)

Pairwise Comparison

- Within concept categories, compare all question pairs
- **1,598 pairs** analyzed (CPS-ACS, FoodAPS-ACS)
- Naive but exhaustive — expect many F3, looking for matches

Multi-Model Ensemble

Three frontier models to reduce bias:

Model	Behavior
OpenAI	Moderate synthesis, slight pro-self bias
Anthropic	High synthesis, neutral
Google	Deferential, conservative

Different profiles = documented finding, not noise

Agreement & Arbitration

- $= \mathbf{0.845}$ — high inter-rater agreement
- ALL pairs sent to arbitration (not just disagreements)
- Enables full behavioral analysis + final verdict

Evaluating AI for Survey Harmonization

Construct Validity: - Three different architectures (OpenAI, Anthropic, Google) with different training - **High convergence ($= 0.845$)** $=$ task is well-defined, not model artifact

Documented Behavioral Differences: - Google: Deferential, conservative (low synthesis) - OpenAI: Moderate synthesis, slight self-bias - Anthropic: High synthesis, neutral

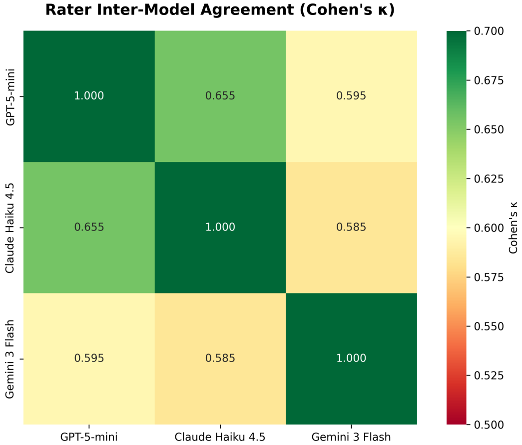
Cost/Quality Design: Fast models rate $\rightarrow$ flagship models arbitrate

Result: Reusable methodology for future survey harmonization work

Construct Validity: Why Trust These Results?

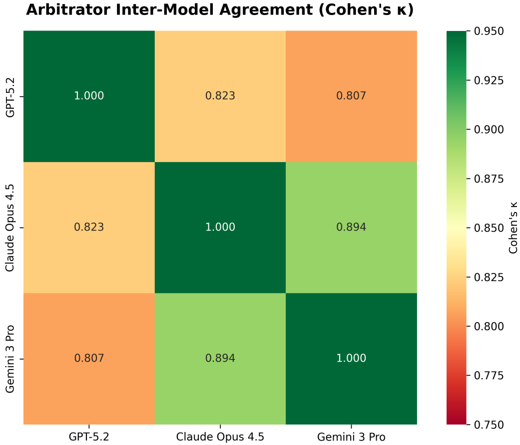
Three independent architectures $\rightarrow$ same judgments $=$ well-defined task

Rater Agreement



Fleiss $= 0.611$ (substantial)

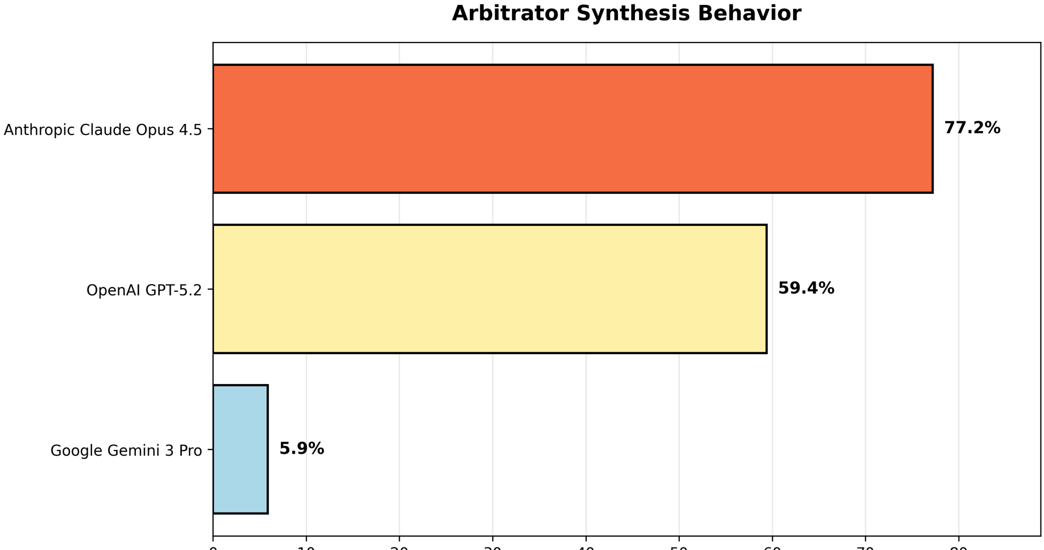
Arbitrator Agreement



Fleiss $= 0.843$ (almost perfect)

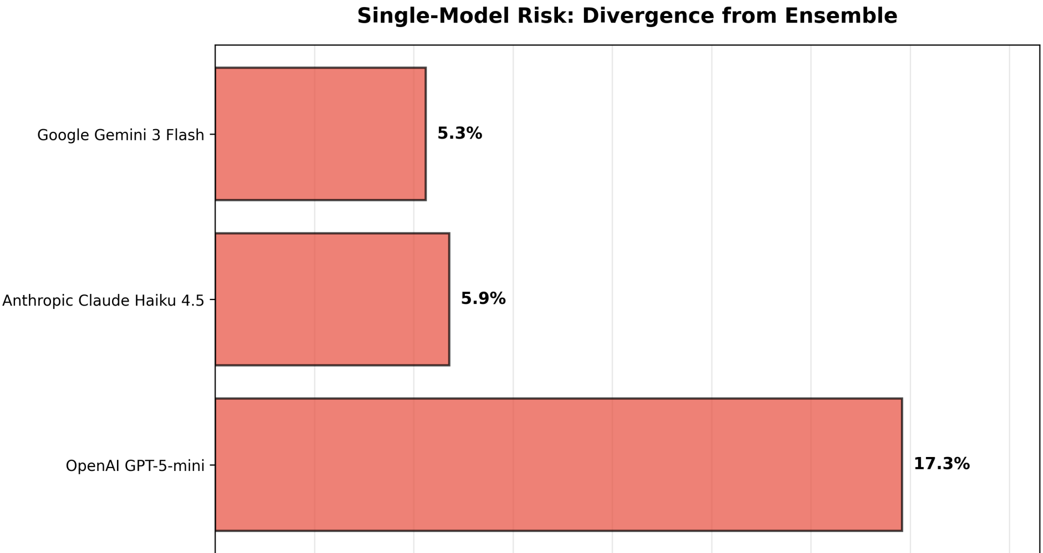
Arbitrator Behavioral Differences

Not a bug — a feature. Diversity captures multiple valid interpretations.

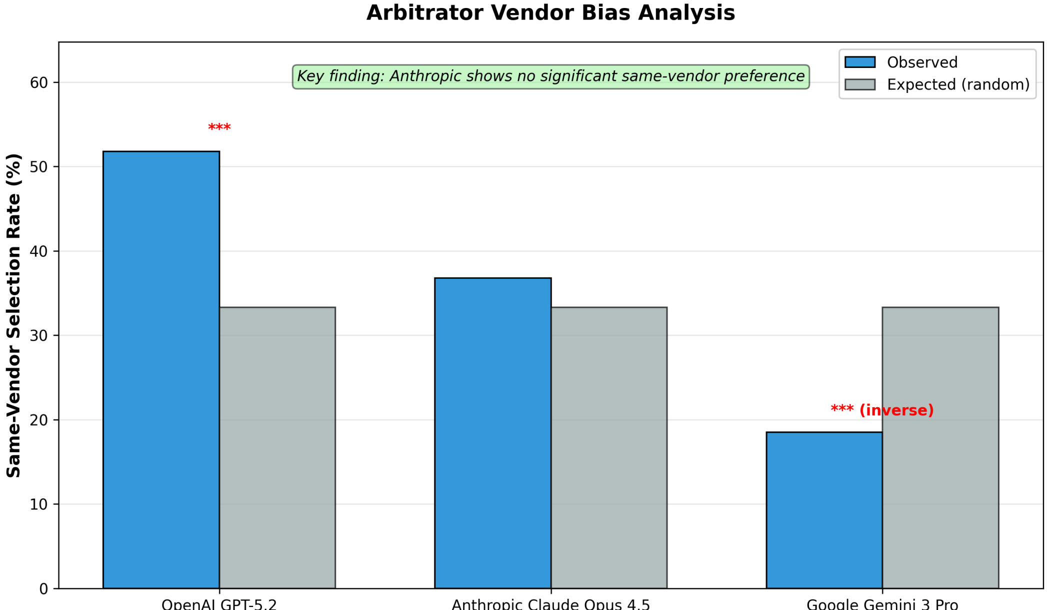


Why Not Just Use One Model?

Single-model divergence from ensemble majority:



Do Arbitrators Favor Their Own Vendor's Rater?



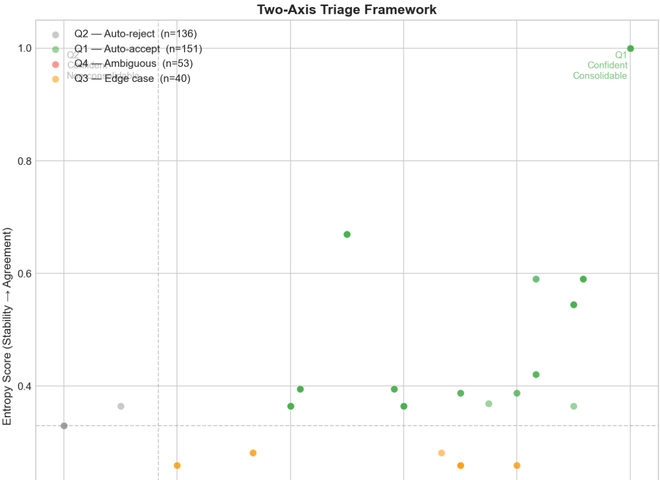
Methodology Validation Summary

Validation Check	Result	Interpretation
Rater convergence	$= 0.611$	Substantial — task well-defined
Arbitrator convergence	$= 0.843$	Almost perfect — disagreements resolvable
Single-model risk	5-17%	Multi-model approach justified
Vendor bias	Mixed	Anthropic unbiased; OpenAI self-favoring

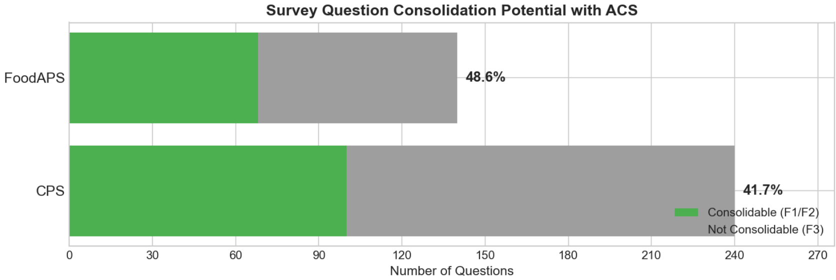
Conclusion: Multi-model ensemble provides robust classifications suitable for survey harmonization analysis.

Question-Level Rollup

- Pair analysis → **380 unique source questions**
- Each with best ACS match
- Two-axis triage: Direction (Borda) × Stability (Entropy)

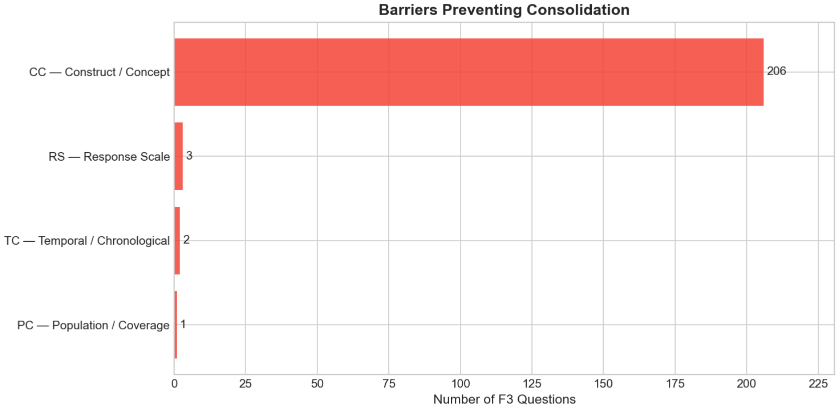


Headline Results



- **CPS:** 41.7% consolidable (100 of 240 questions)
- **FoodAPS:** 48.6% consolidable (68 of 140 questions)

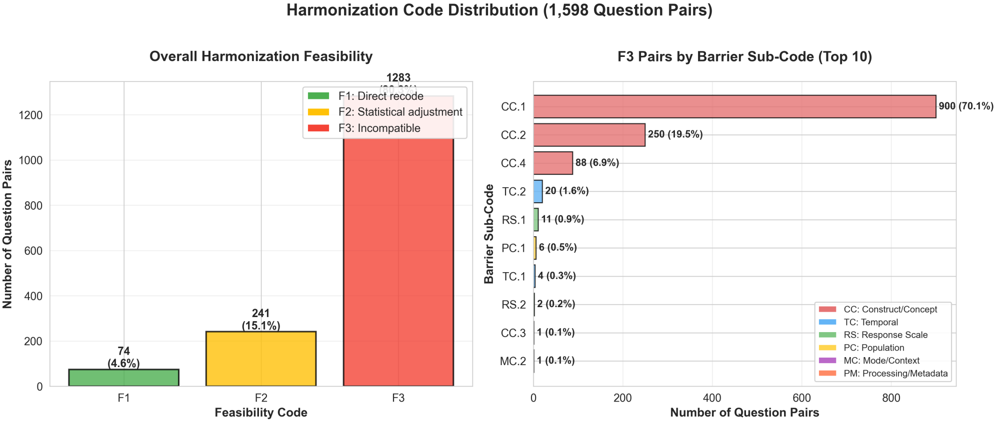
Why Questions Can't Consolidate



97% of F3 = Construct/Concept differences (CC)

Not fixable barriers — fundamentally different measurements

Harmonization Code Distribution



Example: High Consolidability (F1)

CPS: "How many hours per week do you USUALLY work?"

ACS: "What is your best estimate of hours per week you usually work?"

- Verdict: **F1** (directly consolidable)
- Confidence: Borda=1.0, Entropy=1.0
- Same construct, same framing, same reference

Example: Medium Consolidability (F2)

CPS: "EXCLUDING overtime, tips, commissions — what is hourly rate?"

ACS: "What is your hourly rate of pay?"

- Verdict: **F2** (consolidable with transformation)
- Transformation needed: scope alignment (ACS includes all components)

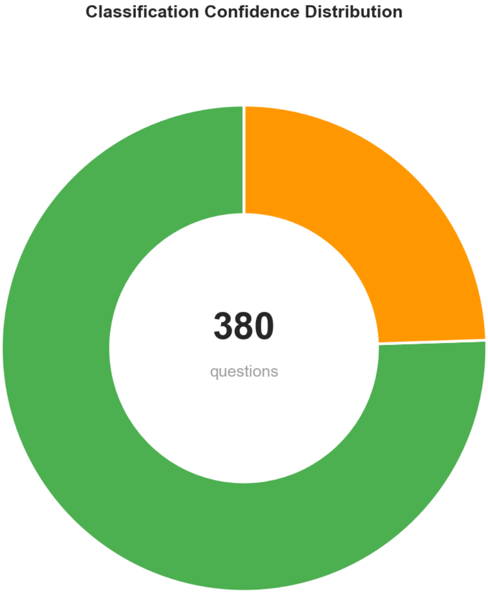
Example: Not Consolidable (F3)

CPS: "Do you WANT to work full time (35+ hours)?"

ACS: "Did this person WORK for pay last week?"

- Verdict: **F3** — Barrier: **CC.1**
- Same domain (employment), entirely different constructs
- Preference vs. behavior — not reconcilable

Expert Review Load



Deliverables

Primary: Expert review table (`expert_review_combined.csv`) - 380 questions with best ACS matches - F1/F2/F3 classifications with confidence scores - Barrier codes and reasoning for each decision - Triage routing (Q1-Q4) for prioritized review

Breakdown: - 168 consolidable (F1+F2) — ready for validation - 212 non-consolidable (F3) — with documented barriers - 93 flagged for expert review — highest-value decisions - Full methodology — reproducible for additional surveys

What This Proves

- $<$ **1 day processing time** for work that traditionally takes weeks/months
- **Expert efficiency $\rightarrow$ faster burden reduction** — identify consolidation opportunities at scale
- **Human oversight preserved throughout** — AI provides initial judgment, experts validate
- **The table is the deliverable** — 380 questions, pre-analyzed, ready for expert review

What's Next

1. Expert validation of recommendations
2. Expand to additional federal surveys
3. Application to survey editing/imputation

Summary

Findings: - $\sim$**44% consolidation potential** — 168 questions could reduce respondent burden - **97% of failures** = construct differences (not fixable)

Method: - **< 1 day processing** for work that traditionally takes weeks/months - **AI provides initial judgment, experts provide final validation** - **The real product:** 380 questions, pre-analyzed, ready for expert review

Impact: Accelerates path to burden reduction through expert efficiency

Questions?

Contact: [your info]

Repository: [link]

Expert review tables: [link to output]

Appendix

Cost/Quality: Is the 3rd Model Worth It?

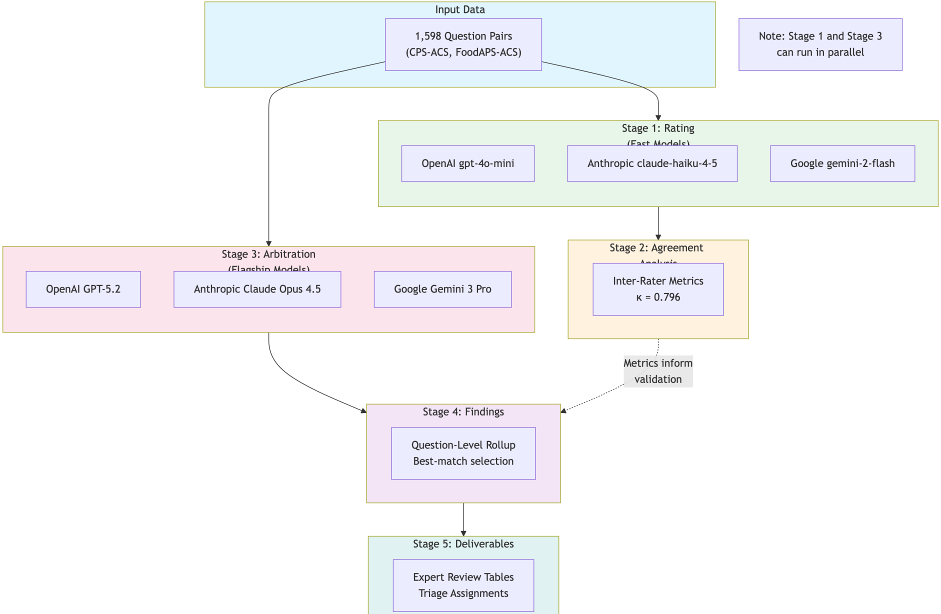
Two-model scenarios — tie frequency:

Pair	Feasibility Ties
OpenAI + Anthropic	23.2%
OpenAI + Google	22.1%
Anthropic + Google	11.0% ← lowest

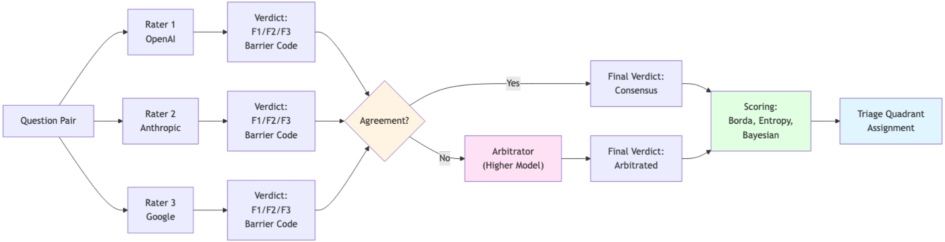
Recommendation: - <1,000 pairs: 2 models + human tiebreaking - >1,000 pairs: 3 models for automated triage

The 3rd model's value is highest when first two disagree.

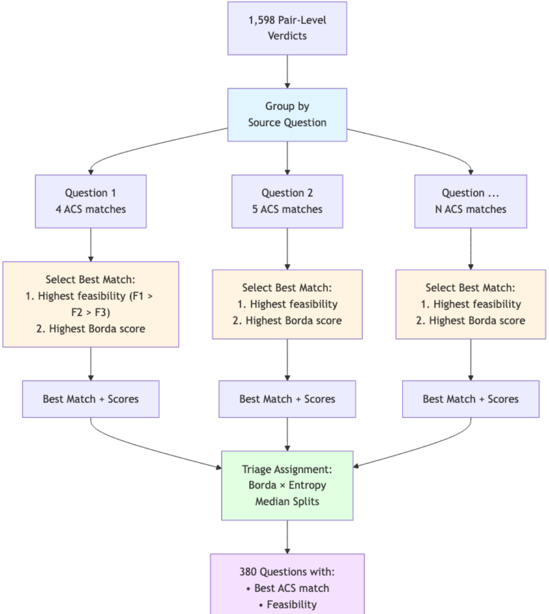
Pipeline Architecture



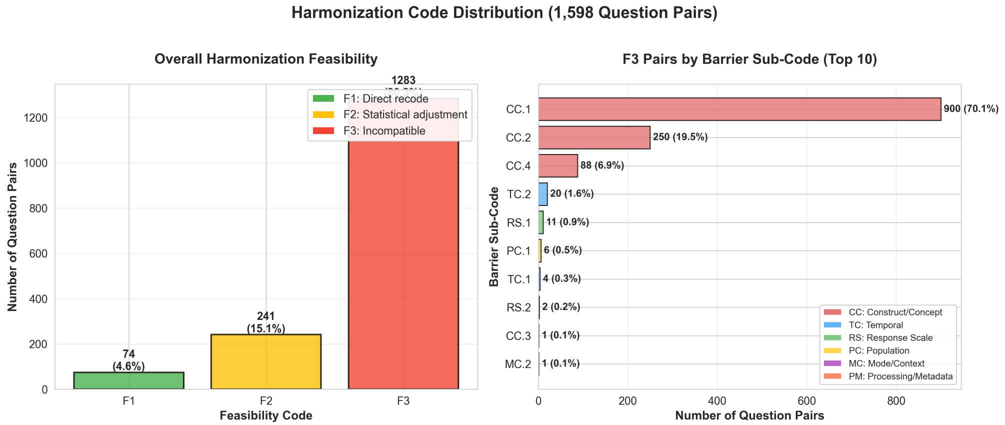
LLM Ensemble Pattern



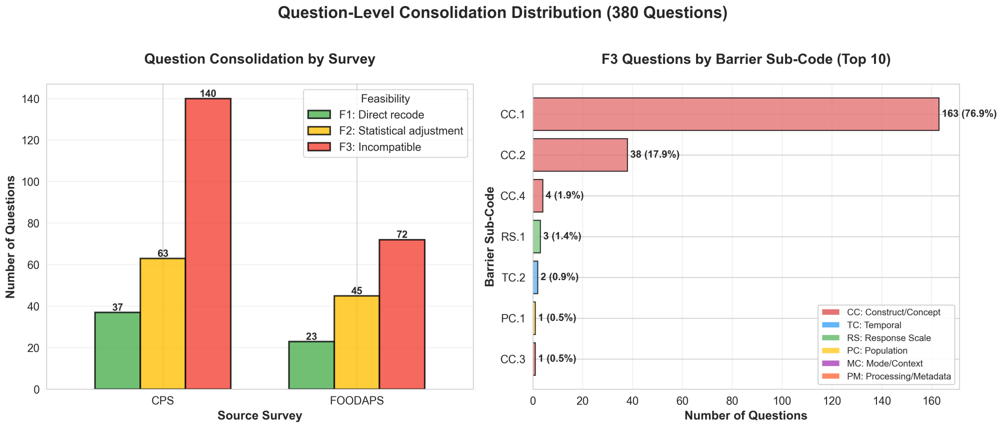
Question-Level Rollup



Harmonization Code Distribution



Question-Level Consolidation Distribution



Harmonization Taxonomy

Primary Source: Fortier et al. (2011, 2017) DataSHaPER/Maelstrom framework

Supporting: Wolf et al. (2016), Saris & Gallhofer (2014), Slomczynski & Tomescu-Dubrow (2018)

Code	Feasibility	Definition	Action Required
F1	**Direct recode**	Mechanically transformable with simple recoding or collapsing	Simple data transformation
F2	**Statistical adjustment**	Requires modeling, imputation, or bridging studies	Statistical harmonization
F3	**Incompatible**	Fundamentally different constructs; not harmonizable without re-fielding	No harmonization possible

Barrier Code Taxonomy (1/3)

Temporal Constraints (TC)

Code	Subtype	Definition	Example
TC	Temporal/Chronological	Reference period or timing differences	—
TC.1	Reference period length	Different recall windows	7-day vs 12-month
TC.2	Temporal framing	Point-in-time vs habitual vs retrospective	“last week” vs “usually”
TC.3	Calendar alignment	Fixed vs rolling reference periods	Calendar year vs “past 12 months”

Construct Constraints (CC)

Code	Subtype	Definition	Example

Barrier Code Taxonomy (2/3)

Population/Coverage Constraints (PC)

Code	Subtype	Definition	Example
PC	Population/Cover-age	Universe, frame, or sample design differences	—
PC.1	Universe definition	Target population differs	Civilian vs all residents
PC.2	Frame exclusions	Different sampling exclusions	Group quarters, territories
PC.3	Age bounds	Different age eligibility	15+ vs 16+ vs 18+
PC.4	Geographic scope	Different geographic coverage	50 states vs states + territories

Response Scale Constraints (RS)

Code	Subtype	Definition	Example

Barrier Code Taxonomy (3/3)

Mode/Context Constraints (MC)

Code	Subtype	Definition	Example
MC	Mode/Context	Interview mode or questionnaire context differences	—
MC.1	Interview mode	Different data collection modes	CATI vs web self-administered
MC.2	Question routing	Different skip patterns	Subset vs all respondents
MC.3	Contextual priming	Preceding questions affect interpretation	Question order effects
MC.4	Proxy response	Proxy vs self-report rules	Household head vs individual

Processing/Metadata Constraints (PM)

Triage Quadrants: Operational Framework

Two-axis classification using Borda (direction) and Entropy (stability):

Quadrant	Borda	Entropy	Count	Description	Action
Q1	High	High	151	Confident consolidable — classifiers agreed, leaning accept	Auto-process, spot-check
Q2	Low	High	136	Confident non-consolidable — classifiers agreed, leaning reject	Auto-process, low priority
Q3	High	Low	40	Uncertain accept — leaning yes but contested	**Expert review priority**
Q4	Low	Low	53	Uncertain reject — genuinely	Expert review

Key Citations

Harmonization Framework:

- Fortier, I., et al. (2011). Is rigorous retrospective harmonization possible? Application of the DataSHaPER approach across 53 large studies. *International Journal of Epidemiology*, 40(5), 1314-1328.
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- Saris, W. E., & Gallhofer, I. N. (2014). *Design, Evaluation, and Analysis of Questionnaires for Survey Research* (2nd ed.). Wiley.

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